

# Test Setup and Performance Evaluation of an Infrared Sensor

## Abstract

This paper describes the feasibility of employing a commercial off-the-shelf (COTS) long-wave infrared (LWIR) imager for satellite-based wildfire detection from low-Earth orbit (LEO) within the SeeSat e.V. ERWIN mission. Using the Teledyne FLIR Boson+ microbolometer as candidate payload sensor, the study combines radiometric simulation and experimental characterization to assess its optical compatibility with the mission's wildfire detection requirements.

In the radiometric analysis, pixel-level received power is computed for a standard 5-lens satellite optics configuration, assuming nadir viewing from a 600 km orbit over a 1000 °C wildfire target. The simulation pipeline incorporates MODTRAN-based atmospheric transmission, the Boson's detector characteristics, and a noise model that includes albedo and glint contributions at the scene level, as well as detector noise represented via the noise equivalent power (NEP) derived from the sensor's datasheet noise equivalent temperature difference (NETD). Under these conditions, fires with an effective area of 55.6 m<sup>2</sup> and larger are found to exceed the detection threshold, providing a conservative lower bound on detectable hotspot size for the assumed geometry and observing conditions.

Complementary laboratory experiments investigate the Boson's thermal sensitivity through NETD measurements. The simulated and measured performance supports the conclusion that the Teledyne FLIR Boson is optically compatible with the ERWIN wildfire detection concept and can serve as a viable COTS LWIR imager for the mission.